pandas3

September 26, 2025

0.0.1 Pandas 3

[1]:
```
import pandas as pd
import numpy as np
```

[4]:
```
movies = pd.read_csv('movies.csv', index_col = 0)
directors = pd.read_csv('directors.csv', index_col = 0)
```

[5]:
```
movies.head()
```

[5]:
```
      id     budget  popularity     revenue  \
0  43597  237000000         150  2787965087
1  43598  300000000         139   961000000
2  43599  245000000         107   880674609
3  43600  250000000         112  1084939099
5  43602  258000000         115   890871626

                                      title  vote_average  vote_count  \
0                                    Avatar           7.2       11800
1  Pirates of the Caribbean: At World's End           6.9        4500
2                                   Spectre           6.3        4466
3                     The Dark Knight Rises           7.6        9106
5                              Spider-Man 3           5.9        3576

   director_id  year month       day
0         4762  2009   Dec  Thursday
1         4763  2007   May  Saturday
2         4764  2015   Oct    Monday
3         4765  2012   Jul    Monday
5         4767  2007   May   Tuesday
```

[7]:
```
directors.head()
```

[7]:
```
       director_name    id gender
0      James Cameron  4762   Male
1     Gore Verbinski  4763   Male
2         Sam Mendes  4764   Male
3  Christopher Nolan  4765   Male
```

```
4     Andrew Stanton  4766   Male
```

[8]:
```
movies.shape
```

[8]:
```
(1465, 11)
```

[9]:
```
directors.shape
```

[9]:
```
(2349, 3)
```

[10]:
```
# How many unique directors have directed movies?
```

[12]:
```
movies['director_id'].nunique()
```

[12]:
```
199
```

[13]:
```
directors['id'].nunique()
```

[13]:
```
2349
```

[21]:
```
# membership operator isin()
```

[16]:
```
'fun' in 'funtooosh'
```

[16]:
```
True
```

[19]:
```
21 in [12, 23, 34, 21]
```

[19]:
```
True
```

[20]:
```
[21, 12] in [21, 12, 13]
```

[20]:
```
False
```

[22]:
```
movies['director_id'].isin(directors['id'])
```

[22]:
```
0       True
1       True
2       True
3       True
5       True
        …
4736    True
4743    True
4748    True
4749    True
4768    True
```

```
Name: director_id, Length: 1465, dtype: bool
```

[23]:
```python
np.all(movies['director_id'].isin(directors['id']))
```

[23]:
```
True
```

[24]:
```python
sum(movies['director_id'].isin(directors['id']))
```

[24]:
```
1465
```

Joining these dataframes:

[28]:
```python
data = movies.merge(directors, how='left', left_on = 'director_id',␣
  ↪right_on='id')
```

[29]:
```python
data.head()
```

[29]:
```
    id_x     budget  popularity     revenue  \
0  43597  237000000         150  2787965087
1  43598  300000000         139   961000000
2  43599  245000000         107   880674609
3  43600  250000000         112  1084939099
4  43602  258000000         115   890871626

                                      title  vote_average  vote_count  \
0                                    Avatar           7.2       11800
1  Pirates of the Caribbean: At World's End           6.9        4500
2                                   Spectre           6.3        4466
3                     The Dark Knight Rises           7.6        9106
4                              Spider-Man 3           5.9        3576

   director_id  year month       day      director_name  id_y gender
0         4762  2009   Dec  Thursday      James Cameron  4762   Male
1         4763  2007   May  Saturday     Gore Verbinski  4763   Male
2         4764  2015   Oct    Monday         Sam Mendes  4764   Male
3         4765  2012   Jul    Monday  Christopher Nolan  4765   Male
4         4767  2007   May   Tuesday          Sam Raimi  4767   Male
```

[34]:
```python
data.drop(['id_x', 'id_y', 'director_id'], axis = 1, inplace=True)
```

[35]:
```python
data.head()
```

[35]:
```
      budget  popularity     revenue  \
0  237000000         150  2787965087
1  300000000         139   961000000
2  245000000         107   880674609
3  250000000         112  1084939099
```

```
4  258000000         115   890871626

                                     title  vote_average  vote_count  year  \
0                                   Avatar           7.2       11800  2009
1  Pirates of the Caribbean: At World's End           6.9        4500  2007
2                                  Spectre           6.3        4466  2015
3                    The Dark Knight Rises           7.6        9106  2012
4                              Spider-Man 3           5.9        3576  2007

  month       day      director_name gender
0   Dec  Thursday      James Cameron   Male
1   May  Saturday     Gore Verbinski   Male
2   Oct    Monday         Sam Mendes   Male
3   Jul    Monday  Christopher Nolan   Male
4   May   Tuesday          Sam Raimi   Male
```

[38]:
```
# Post read: https://colab.research.google.com/drive/
↪1yrfHSQYUMxxLKGUG-gCPf-R232BuimiR?usp=sharing
```

[44]:
```
# Quiz: How to filter those records where movies are released either in the␣
↪year
# 2015, 2016 or 2012 from the movies dataset?


# df['year'].isin([2015, 2016, 2012])

# df['year'].in([2015, 2016, 2012])

# df['year']==([2015, 2016, 2012])
```

[ ]:
```

```

0.0.2 apply()

- Similar to vectorization in numpy. It helps us to apply a function on all values of a df/series

[45]:
```
data.head()
```

[45]:
```
      budget  popularity     revenue  \
0  237000000         150  2787965087
1  300000000         139   961000000
2  245000000         107   880674609
3  250000000         112  1084939099
4  258000000         115   890871626

                                     title  vote_average  vote_count  year  \
0                                   Avatar           7.2       11800  2009
```

```
1  Pirates of the Caribbean: At World's End          6.9         4500  2007
2                                   Spectre          6.3         4466  2015
3                     The Dark Knight Rises          7.6         9106  2012
4                              Spider-Man 3          5.9         3576  2007

  month       day      director_name gender
0   Dec  Thursday      James Cameron   Male
1   May  Saturday     Gore Verbinski   Male
2   Oct    Monday         Sam Mendes   Male
3   Jul    Monday  Christopher Nolan   Male
4   May   Tuesday          Sam Raimi   Male
```

[48]:
```python
def encode(x):
    if x == "Male":
        return 0
    else:
        return 1
```

[49]:
```python
encode('Male')
```

[49]: 0

[50]:
```python
encode("Female")
```

[50]: 1

[52]:
```python
data['gender'] = data['gender'].apply(encode)
```

[54]:
```python
data.head()
```

[54]:
```
      budget  popularity     revenue  \
0  237000000         150  2787965087
1  300000000         139   961000000
2  245000000         107   880674609
3  250000000         112  1084939099
4  258000000         115   890871626

                                      title  vote_average  vote_count  year  \
0                                    Avatar           7.2       11800  2009
1  Pirates of the Caribbean: At World's End           6.9        4500  2007
2                                   Spectre           6.3        4466  2015
3                     The Dark Knight Rises           7.6        9106  2012
4                              Spider-Man 3           5.9        3576  2007

  month       day      director_name  gender
0   Dec  Thursday      James Cameron       0
1   May  Saturday     Gore Verbinski       0
```

```
2   Oct    Monday          Sam Mendes       0
3   Jul    Monday  Christopher Nolan       0
4   May   Tuesday           Sam Raimi       0
```

[55]:
```
# Can we add 'revenue' and 'budget' columns to get total transaction for each
↪movies
```

[56]:
```
data[['revenue', 'budget']].apply(np.sum)
```

[56]:
```
revenue    209866997305
budget      70353617179
dtype: int64
```

[61]:
```
data['transaction'] = data[['revenue', 'budget']].apply(np.sum, axis = 1)
```

[62]:
```
data.head()
```

[62]:
```
      budget  popularity     revenue  \
0  237000000         150  2787965087
1  300000000         139   961000000
2  245000000         107   880674609
3  250000000         112  1084939099
4  258000000         115   890871626

                                      title  vote_average  vote_count  year  \
0                                    Avatar           7.2       11800  2009
1  Pirates of the Caribbean: At World's End           6.9        4500  2007
2                                   Spectre           6.3        4466  2015
3                     The Dark Knight Rises           7.6        9106  2012
4                              Spider-Man 3           5.9        3576  2007

  month       day      director_name  gender  transaction
0   Dec  Thursday      James Cameron       0   3024965087
1   May  Saturday     Gore Verbinski       0   1261000000
2   Oct    Monday         Sam Mendes       0   1125674609
3   Jul    Monday  Christopher Nolan       0   1334939099
4   May   Tuesday          Sam Raimi       0   1148871626
```

[70]:
```
data['revenue'] - data['budget']
```

[70]:
```
0       2550965087
1        661000000
2        635674609
3        834939099
4        632871626
           …
1460        321952
```

```
1461       3124130
1462             0
1463             0
1464       1820920
Length: 1465, dtype: int64
```

[63]:
```
# Get total profit per movie?
```

[67]:
```
def prof(row):
    return row['revenue'] - row['budget']
```

[68]:
```
data['profit'] = data[['revenue', 'budget']].apply(prof, axis = 1)
```

[69]:
```
data.head()
```

[69]:
```
      budget  popularity     revenue  \
0  237000000         150  2787965087
1  300000000         139   961000000
2  245000000         107   880674609
3  250000000         112  1084939099
4  258000000         115   890871626

                                      title  vote_average  vote_count  year  \
0                                    Avatar           7.2       11800  2009
1  Pirates of the Caribbean: At World's End           6.9        4500  2007
2                                   Spectre           6.3        4466  2015
3                     The Dark Knight Rises           7.6        9106  2012
4                              Spider-Man 3           5.9        3576  2007

  month       day      director_name  gender  transaction      profit
0   Dec  Thursday      James Cameron       0   3024965087  2550965087
1   May  Saturday     Gore Verbinski       0   1261000000   661000000
2   Oct    Monday         Sam Mendes       0   1125674609   635674609
3   Jul    Monday  Christopher Nolan       0   1334939099   834939099
4   May   Tuesday          Sam Raimi       0   1148871626   632871626
```

[ ]:

0.0.3 Grouping of data

[71]:
```
# Get data director wise
```

[72]:
```
data.head()
```

[72]:
```
      budget  popularity     revenue  \
0  237000000         150  2787965087
1  300000000         139   961000000
```

```
2  245000000          107   880674609
3  250000000          112  1084939099
4  258000000          115   890871626

                                      title  vote_average  vote_count  year  \
0                                    Avatar           7.2       11800  2009
1  Pirates of the Caribbean: At World's End           6.9        4500  2007
2                                   Spectre           6.3        4466  2015
3                     The Dark Knight Rises           7.6        9106  2012
4                              Spider-Man 3           5.9        3576  2007

  month       day      director_name  gender  transaction      profit
0   Dec  Thursday      James Cameron       0   3024965087  2550965087
1   May  Saturday     Gore Verbinski       0   1261000000   661000000
2   Oct    Monday         Sam Mendes       0   1125674609   635674609
3   Jul    Monday  Christopher Nolan       0   1334939099   834939099
4   May   Tuesday          Sam Raimi       0   1148871626   632871626
```

[73]:
```python
data.groupby('director_name')
```

[73]: <pandas.core.groupby.generic.DataFrameGroupBy object at 0x7fe9f93c76a0>

[74]:
```python
# What's the number of groups our data is divided into
```

[75]:
```python
data['director_name'].nunique()
```

[75]: 199

[76]:
```python
data.groupby('director_name').ngroups
```

[76]: 199

[77]:
```python
data.groupby('director_name').groups
```

[77]:
```
{'Adam McKay': [176, 323, 366, 505, 839, 916], 'Adam Shankman': [265, 300, 350,
404, 458, 843, 999, 1231], 'Alejandro González Iñárritu': [106, 749, 1015, 1034,
1077, 1405], 'Alex Proyas': [95, 159, 514, 671, 873], 'Alexander Payne': [793,
1006, 1101, 1211, 1281], 'Andrew Adamson': [11, 43, 328, 501, 947], 'Andrew
Niccol': [533, 603, 701, 722, 1439], 'Andrzej Bartkowiak': [349, 549, 754, 911,
924], 'Andy Fickman': [517, 681, 909, 926, 973, 1023], 'Andy Tennant': [314,
320, 464, 593, 676, 885], 'Ang Lee': [99, 134, 748, 840, 1089, 1110, 1132,
1184], 'Anne Fletcher': [610, 650, 736, 789, 1206], 'Antoine Fuqua': [310, 338,
424, 467, 576, 808, 818, 1105], 'Atom Egoyan': [946, 1128, 1164, 1194, 1347,
1416], 'Barry Levinson': [313, 319, 471, 594, 878, 898, 1013, 1037, 1082, 1143,
1185, 1345, 1378], 'Barry Sonnenfeld': [13, 48, 90, 205, 591, 778, 783], 'Ben
Stiller': [209, 212, 547, 562, 850], 'Bill Condon': [102, 307, 902, 1233, 1381],
'Bobby Farrelly': [352, 356, 481, 498, 624, 630, 654, 806, 928, 972, 1111],
```

'Brad Anderson': [1163, 1197, 1350, 1419, 1430], 'Brett Ratner': [24, 39, 188, 207, 238, 292, 405, 456, 920], 'Brian De Palma': [228, 255, 318, 439, 747, 905, 919, 1088, 1232, 1261, 1317, 1354], 'Brian Helgeland': [512, 607, 623, 742, 933], 'Brian Levant': [418, 449, 568, 761, 860, 1003], 'Brian Robbins': [416, 441, 669, 962, 988, 1115], 'Bryan Singer': [6, 32, 33, 44, 122, 216, 297, 1326], 'Cameron Crowe': [335, 434, 488, 503, 513, 698], 'Catherine Hardwicke': [602, 695, 724, 937, 1406, 1412], 'Chris Columbus': [117, 167, 204, 218, 229, 509, 656, 897, 996, 1086, 1129], 'Chris Weitz': [17, 500, 794, 869, 1202, 1267], 'Christopher Nolan': [3, 45, 58, 59, 74, 565, 641, 1341], 'Chuck Russell': [177, 410, 657, 1069, 1097, 1339], 'Clint Eastwood': [369, 426, 447, 482, 490, 520, 530, 535, 645, 727, 731, 786, 787, 899, 974, 986, 1167, 1190, 1313], 'Curtis Hanson': [494, 579, 606, 711, 733, 1057, 1310], 'Danny Boyle': [527, 668, 1083, 1085, 1126, 1168, 1287, 1385], 'Darren Aronofsky': [113, 751, 1187, 1328, 1363, 1458], 'Darren Lynn Bousman': [1241, 1243, 1283, 1338, 1440], 'David Ayer': [50, 273, 741, 1024, 1146, 1407], 'David Cronenberg': [541, 767, 994, 1055, 1254, 1268, 1334], 'David Fincher': [62, 213, 253, 383, 398, 478, 522, 555, 618, 785], 'David Gordon Green': [543, 862, 884, 927, 1376, 1418, 1432, 1459], 'David Koepp': [443, 644, 735, 1041, 1209], 'David Lynch': [583, 1161, 1264, 1340, 1456], 'David O. Russell': [422, 556, 609, 896, 982, 989, 1229, 1304], 'David R. Ellis': [582, 634, 756, 888, 934], 'David Zucker': [569, 619, 965, 1052, 1175], 'Dennis Dugan': [217, 260, 267, 293, 303, 718, 780, 977, 1247], 'Donald Petrie': [427, 507, 570, 649, 858, 894, 1106, 1331], 'Doug Liman': [52, 148, 251, 399, 544, 1318, 1451], 'Edward Zwick': [92, 182, 346, 566, 791, 819, 825], 'F. Gary Gray': [308, 402, 491, 523, 697, 833, 1272, 1380], 'Francis Ford Coppola': [487, 559, 622, 646, 772, 1076, 1155, 1253, 1312], 'Francis Lawrence': [63, 72, 109, 120, 679], 'Frank Coraci': [157, 249, 275, 451, 577, 599, 963], 'Frank Oz': [193, 355, 473, 580, 712, 813, 987], 'Garry Marshall': [329, 496, 528, 571, 784, 893, 1029, 1169], 'Gary Fleder': [518, 667, 689, 867, 981, 1165], 'Gary Winick': [258, 797, 798, 804, 1454], 'Gavin O'Connor': [820, 841, 939, 953, 1444], 'George A. Romero': [250, 1066, 1096, 1278, 1367, 1396], 'George Clooney': [343, 450, 831, 966, 1302], 'George Miller': [78, 103, 233, 287, 1250, 1403, 1450], 'Gore Verbinski': [1, 8, 9, 107, 119, 633, 1040], 'Guillermo del Toro': [35, 252, 419, 486, 1118], 'Gus Van Sant': [595, 1018, 1027, 1159, 1240, 1311, 1398], 'Guy Ritchie': [124, 215, 312, 1093, 1225, 1269, 1420], 'Harold Ramis': [425, 431, 558, 586, 788, 1137, 1166, 1325], 'Ivan Reitman': [274, 643, 816, 883, 910, 935, 1134, 1242], 'James Cameron': [0, 19, 170, 173, 344, 1100, 1320], 'James Ivory': [1125, 1152, 1180, 1291, 1293, 1390, 1397], 'James Mangold': [140, 141, 557, 560, 829, 845, 958, 1145], 'James Wan': [30, 617, 1002, 1047, 1337, 1417, 1424], 'Jan de Bont': [155, 224, 231, 270, 781], 'Jason Friedberg': [812, 1010, 1012, 1014, 1036], 'Jason Reitman': [792, 1092, 1213, 1295, 1299], 'Jaume Collet-Serra': [516, 540, 640, 725, 1011, 1189], 'Jay Roach': [195, 359, 389, 397, 461, 703, 859, 1072], 'Jean-Pierre Jeunet': [423, 485, 605, 664, 765], 'Joe Dante': [284, 525, 638, 1226, 1298, 1428], 'Joe Wright': [85, 432, 553, 803, 814, 855], 'Joel Coen': [428, 670, 691, 707, 721, 889, 906, 980, 1157, 1238, 1305], 'Joel Schumacher': [128, 184, 348, 484, 572, 614, 652, 764, 876, 886, 1108, 1230, 1280], 'John Carpenter': [537, 663, 686, 861, 938, 1028, 1080, 1102, 1329, 1371], 'John Glen': [601, 642, 801, 847, 864], 'John Landis': [524, 868,

```
1276, 1384, 1435], 'John Madden': [457, 882, 1020, 1249, 1257], 'John
McTiernan': [127, 214, 244, 351, 534, 563, 648, 782, 838, 1074], 'John
Singleton': [294, 489, 732, 796, 1120, 1173, 1316], 'John Whitesell': [499, 632,
763, 1119, 1148], 'John Woo': [131, 142, 264, 371, 420, 675, 1182], 'Jon
Favreau': [46, 54, 55, 382, 759, 1346], 'Jon M. Chu': [100, 225, 810, 1099,
1186], 'Jon Turteltaub': [64, 180, 372, 480, 760, 846, 1171], 'Jonathan Demme':
[277, 493, 1000, 1123, 1215], 'Jonathan Liebesman': [81, 143, 339, 1117, 1301],
'Judd Apatow': [321, 710, 717, 865, 881], 'Justin Lin': [38, 123, 246, 1437,
1447], 'Kenneth Branagh': [80, 197, 421, 879, 1094, 1277, 1288], 'Kenny Ortega':
[412, 852, 1228, 1315, 1365], 'Kevin Reynolds': [53, 502, 639, 1019, 1059], …}
```

[78]:
```
# Let's extract data of a particular director from this grouped data?
```

[79]:
```
data.groupby('director_name').get_group('Christopher Nolan')
```

[79]:
```
         budget  popularity     revenue                  title  vote_average  \
3     250000000         112  1084939099  The Dark Knight Rises           7.6
45    185000000         187  1004558444        The Dark Knight           8.2
58    165000000         724   675120017           Interstellar           8.1
59    160000000         167   825532764              Inception           8.1
74    150000000         115   374218673          Batman Begins           7.5
565    46000000          41   113714830               Insomnia           6.8
641    40000000          74   109676311           The Prestige           8.0
1341    9000000          60    39723096                Memento           8.1

      vote_count  year month        day      director_name  gender  \
3           9106  2012   Jul     Monday  Christopher Nolan       0
45         12002  2008   Jul  Wednesday  Christopher Nolan       0
58         10867  2014   Nov  Wednesday  Christopher Nolan       0
59         13752  2010   Jul  Wednesday  Christopher Nolan       0
74          7359  2005   Jun     Friday  Christopher Nolan       0
565         1148  2002   May     Friday  Christopher Nolan       0
641         4391  2006   Oct   Thursday  Christopher Nolan       0
1341        4028  2000   Oct  Wednesday  Christopher Nolan       0

      transaction     profit
3      1334939099  834939099
45     1189558444  819558444
58      840120017  510120017
59      985532764  665532764
74      524218673  224218673
565     159714830   67714830
641     149676311   69676311
1341     48723096   30723096
```

[80]:
```
data.groupby('director_name').get_group('James Cameron')
```

[80]:
```
          budget  popularity     revenue                          title  \
0      237000000         150  2787965087                         Avatar
19     200000000         100  1845034188                        Titanic
170    100000000         101   520000000  Terminator 2: Judgment Day
173    115000000          38   378882411                      True Lies
344     70000000          24    90000098                      The Abyss
1100    18500000          67   183316455                         Aliens
1320     6400000          74    78371200                 The Terminator

      vote_average  vote_count  year month        day director_name  gender  \
0              7.2       11800  2009   Dec   Thursday  James Cameron       0
19             7.5        7562  1997   Nov    Tuesday  James Cameron       0
170            7.7        4185  1991   Jul     Monday  James Cameron       0
173            6.8        1116  1994   Jul   Thursday  James Cameron       0
344            7.1         808  1989   Aug  Wednesday  James Cameron       0
1100           7.7        3220  1986   Jul     Friday  James Cameron       0
1320           7.3        4128  1984   Oct     Friday  James Cameron       0

      transaction      profit
0      3024965087  2550965087
19     2045034188  1645034188
170     620000000   420000000
173     493882411   263882411
344     160000098    20000098
1100    201816455   164816455
1320     84771200    71971200
```

[81]:
```python
# Find count of movies by each director
```

[85]:
```python
data.groupby('director_name')['title'].count()
```

[85]:
```
director_name
Adam McKay                    6
Adam Shankman                 8
Alejandro González Iñárritu   6
Alex Proyas                   5
Alexander Payne               5
                             ..
Wes Craven                   10
Wolfgang Petersen             7
Woody Allen                  18
Zack Snyder                   7
Zhang Yimou                   6
Name: title, Length: 199, dtype: int64
```

[89]:
```python
data.groupby('director_name')['title'].count().max()
```

[89]: 26

Applying multiple aggregates on a feature

- aggregate
- agg

[91]:
```
# Find min and max year in which a director has directed a movie
```

[94]:
```
data.groupby('director_name')['year'].aggregate(['min', 'max', 'mean'])
```

[94]:
```
                             min   max         mean
director_name
Adam McKay                  2004  2015  2009.333333
Adam Shankman               2001  2012  2005.375000
Alejandro González Iñárritu  2000  2015  2008.000000
Alex Proyas                 1994  2016  2004.200000
Alexander Payne             1999  2013  2005.800000
…                            …     …            …
Wes Craven                  1984  2011  1999.700000
Wolfgang Petersen           1981  2006  1995.285714
Woody Allen                 1977  2013  2001.611111
Zack Snyder                 2004  2016  2009.857143
Zhang Yimou                 2002  2014  2007.666667

[199 rows x 3 columns]
```

[93]:
```
data.groupby('director_name')['year'].agg(['min', 'max'])
```

[93]:
```
                             min   max
director_name
Adam McKay                  2004  2015
Adam Shankman               2001  2012
Alejandro González Iñárritu  2000  2015
Alex Proyas                 1994  2016
Alexander Payne             1999  2013
…                            …     …
Wes Craven                  1984  2011
Wolfgang Petersen           1981  2006
Woody Allen                 1977  2013
Zack Snyder                 2004  2016
Zhang Yimou                 2002  2014

[199 rows x 2 columns]
```

[96]:
```
#  Quiz:
```

[97]:
```
# data.groupby('director_name').mean()
```

[ ]:
```

```

Group based filtering

[100]:
```
# Get details of the movies by high budget directors only => any director with
# atleast 1 movies budget > 10000000
```

[101]:
```
data.head()
```

[101]:
```
      budget  popularity     revenue  \
0  237000000         150  2787965087
1  300000000         139   961000000
2  245000000         107   880674609
3  250000000         112  1084939099
4  258000000         115   890871626

                                      title  vote_average  vote_count  year  \
0                                    Avatar           7.2       11800  2009
1  Pirates of the Caribbean: At World's End           6.9        4500  2007
2                                   Spectre           6.3        4466  2015
3                     The Dark Knight Rises           7.6        9106  2012
4                              Spider-Man 3           5.9        3576  2007

  month       day      director_name  gender  transaction      profit
0   Dec  Thursday      James Cameron       0   3024965087  2550965087
1   May  Saturday     Gore Verbinski       0   1261000000   661000000
2   Oct    Monday         Sam Mendes       0   1125674609   635674609
3   Jul    Monday  Christopher Nolan       0   1334939099   834939099
4   May   Tuesday          Sam Raimi       0   1148871626   632871626
```

[109]:
```
data_dir = data.groupby('director_name')['budget'].max().reset_index()
```

[113]:
```
data_dir['budget'] > 10000000
```

[113]:
```
0       True
1       True
2       True
3       True
4       True
        …
194     True
195     True
196     True
197     True
198     True
```

```
Name: budget, Length: 199, dtype: bool
```

[116]:
```
names = data_dir[data_dir['budget'] > 10000000]['director_name']
```

[119]:
```
data[data['director_name'].isin(names)]
```

[119]:
```
         budget  popularity     revenue  \
0     237000000         150  2787965087
1     300000000         139   961000000
2     245000000         107   880674609
3     250000000         112  1084939099
4     258000000         115   890871626
…           …           …           …
1460          0           3      321952
1461      27000          19     3151130
1462          0           7           0
1463          0           3           0
1464     220000          14     2040920

                                         title  vote_average  vote_count  \
0                                       Avatar           7.2       11800
1     Pirates of the Caribbean: At World's End           6.9        4500
2                                      Spectre           6.3        4466
3                        The Dark Knight Rises           7.6        9106
4                                 Spider-Man 3           5.9        3576
…                                          …           …          …
1460                            The Last Waltz           7.9          64
1461                                    Clerks           7.4         755
1462                                   Rampage           6.0         131
1463                                   Slacker           6.4          77
1464                               El Mariachi           6.6         238

      year month       day       director_name  gender  transaction      profit
0     2009   Dec  Thursday       James Cameron       0   3024965087  2550965087
1     2007   May  Saturday      Gore Verbinski       0   1261000000   661000000
2     2015   Oct    Monday          Sam Mendes       0   1125674609   635674609
3     2012   Jul    Monday   Christopher Nolan       0   1334939099   834939099
4     2007   May   Tuesday           Sam Raimi       0   1148871626   632871626
…      …    …       …                 …     …          …           …
1460  1978   May    Monday     Martin Scorsese       0       321952      321952
1461  1994   Sep   Tuesday         Kevin Smith       0      3178130     3124130
1462  2009   Aug    Friday            Uwe Boll       0            0           0
1463  1990   Jul    Friday   Richard Linklater       0            0           0
1464  1992   Sep    Friday    Robert Rodriguez       1      2260920     1820920

[1455 rows x 13 columns]
```

```
[ ]:
```

```
[121]: # Using filter method in a single line
```

```
[123]: data.groupby('director_name').filter(lambda x : x['budget'].max() > 10000000)
```

```
[123]:         budget  popularity     revenue  \
0      237000000         150  2787965087
1      300000000         139   961000000
2      245000000         107   880674609
3      250000000         112  1084939099
4      258000000         115   890871626
…            …           …           …
1460           0           3      321952
1461       27000          19     3151130
1462           0           7           0
1463           0           3           0
1464      220000          14     2040920

                                          title  vote_average  vote_count  \
0                                        Avatar           7.2       11800
1      Pirates of the Caribbean: At World's End           6.9        4500
2                                       Spectre           6.3        4466
3                         The Dark Knight Rises           7.6        9106
4                                  Spider-Man 3           5.9        3576
…                                           …             …           …
1460                             The Last Waltz           7.9          64
1461                                     Clerks           7.4         755
1462                                    Rampage           6.0         131
1463                                    Slacker           6.4          77
1464                                El Mariachi           6.6         238

      year month       day       director_name  gender  transaction      profit
0     2009   Dec  Thursday       James Cameron       0   3024965087  2550965087
1     2007   May  Saturday      Gore Verbinski       0   1261000000   661000000
2     2015   Oct    Monday          Sam Mendes       0   1125674609   635674609
3     2012   Jul    Monday   Christopher Nolan       0   1334939099   834939099
4     2007   May   Tuesday           Sam Raimi       0   1148871626   632871626
…      …     …         …                   …      …            …           …
1460  1978   May    Monday    Martin Scorsese       0       321952      321952
1461  1994   Sep   Tuesday        Kevin Smith       0      3178130     3124130
1462  2009   Aug    Friday           Uwe Boll       0            0           0
1463  1990   Jul    Friday  Richard Linklater       0            0           0
1464  1992   Sep    Friday   Robert Rodriguez       1      2260920     1820920

[1455 rows x 13 columns]
```

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]:

[ ]: